Sanny Hong

AI Engineer · Production LLM & Agentic Systems · Evaluation · Cost & Latency

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SUMMARY

AI engineer who builds and runs LLM systems in production. In the technical department of an industrial-IoT company, I shipped provider-agnostic agentic assistants over a 19M-row database, cut tokens per question by 84%, and check answers against ground-truth queries on live data. I also ship computer-vision and time-series models that run on CPU-only hardware.

SKILLS

- **LLM & agents:** OpenAI and Anthropic APIs, tool/function calling, agentic loops, context engineering, RAG and grounding, structured outputs, provider-agnostic model layer
- **Evaluation & reliability:** ground-truth evaluation, deterministic tests, error analysis, entity resolution, per-request cost caps, rate-limit (TPM) handling, structured run/step tracing
- **Machine learning:** PyTorch, OpenCV, segmentation, OCR, detection and tracking, anomaly detection (Isolation Forest, LOF, LSTM, S-H-ESD), time-series forecasting, CPU inference
- **Engineering:** Python, SQL, TypeScript · FastAPI, Django, React · PostgreSQL, SQL Server · Docker · ETL, dimensional modelling · CI/CD

EXPERIENCE

Software Engineer · Flotech Controls, Singapore · *Dec 2024 – Present*

Technical department. Builds and runs AI systems for public-sector water-infrastructure clients (SCADA / industrial IoT).

LLM & agentic systems

- Shipped production LLM assistants over a **447-station, 19M-row** time-series database: an agentic tool-calling loop over parameterised, **read-only SQL**, so the model can query but never modify production data.
- Built a **provider-agnostic model layer (OpenAI / Anthropic)**, including a compatibility shim for GPT-5 API changes, so models can be swapped without changing agent code.
- Cut tokens per question (input + output) from **~233k to ~36k (−84%)** through context engineering, tracked as cost per response; this kept traffic under provider rate limits. Enforced **per-request USD cost caps** in production.
- Wrote **ground-truth evaluation scripts** that check assistant answers against direct SQL on live data. They caught a **silent mis-join**: the agent picked join columns by name similarity, linked the wrong columns across tables, and attributed work orders to 1 site instead of 46. Fixed it with a **schema-relationship context file** that tells the agent how the tables link; also added entity resolution for location names.
- Logged every agent step and run as **structured JSONL traces** for failure analysis.

Computer vision & time series

- Flood-level CV pipeline: fine-tuned DeepLabV3-MobileNetV3 (**IoU 0.841**) and fused six detectors with OCR auto-calibration, cutting reading error from **~137px to ~1px**. Re-engineered a Florence-2 + SAM2 GPU pipeline into a **CPU-only OpenCV stack (<1 s/image) with the same accuracy on the same test set**, plus auto-generated training labels.
- Production anomaly detection on high-frequency water-quality data (Isolation Forest, LOF, LSTM, S-H-ESD) with ETL, scheduling and alerting; redesigned flat tables into a star schema for fast SCADA analytics.

Independent Projects & Further Study · Singapore · *Dec 2022 – Nov 2024*

- Built an Amazon-style e-commerce web app with a content-based recommender, later rebuilt as the forecasting engine below.
- Admitted to SMU's Master of IT in Business (AI) on scholarship (2024).

Software Engineer (Frontend Lead) · GienTech (IT services; seconded to OCBC), Shanghai · *Feb 2022 – Dec 2022*

- Led front-end development of OCBC Malaysia's SIBS Connect financial-record system (1,000+ users) across the full lifecycle; mentored junior engineers; SonarQube + Jenkins CI/CD with 90%+ deployment success.
- Delivered an HR management system (React / TypeScript / Django) for 300+ staff on bi-weekly releases. During a JavaScript → TypeScript migration, consolidated duplicated logic into shared React contexts, **cutting the codebase by ~40%**.

PROJECTS

- **Forecasting Engine** · *2023 – Present* · github.com/HSanny/forecasting-engine
  Natural-language agent over a dataset-agnostic recommender and time-series demand forecaster. Retrieval grounds every answer in real catalogue items, and the agent **refuses when nothing matches** instead of inventing products. Each answer carries its intent, the tool used and its evidence. Adapters for H&M and Amazon data; 12 deterministic tests.
- **Motion Analyser** · *2026 – Present* · github.com/HSanny/motion-analyser
  Turns one camera into an event stream: detection + tracking feed line-crossing, zone-dwell and speed analytics. Analytics core unit-tested; end-to-end run on real footage in progress. Public video only.

EDUCATION

Master of IT in Business (Artificial Intelligence) · Singapore Management University · *2024 – Present* · SMU MITB Scholarship

Bachelor of Information Technology · The Australian National University · *2017 – 2022* · Majors: Computer Science & Mathematics · Specialisation: Intelligent Systems

Languages: English (full professional) · Mandarin, Cantonese (native) · Japanese (conversational)